

TOP 50 CASE STUDIES: BIOLOGY DOMINATOR

Short-Form Boards Ready Answer Key • © Thunderstudy Series

By Pragya Thunderstudy

I. Molecular Basis of Inheritance

1. DNA Replication Fork: Leading vs Lagging Strands

Q1. Why is lagging strand synthesized discontinuously?

➤ Because DNA polymerase works only in 5' → 3' direction.

Q2. Name enzyme joining fragments.

➤ DNA ligase.

2. Hershey-Chase Experiment (32P and 35S)

Q1. Why was 32P used?

➤ To label DNA specifically (phosphorus is absent in protein).

Q2. Result/Conclusion?

➤ DNA is the true genetic material.

3. Lac Operon (Lactose Introduction)

Q1. Role of Repressor?

➤ Binds to operator to block RNA polymerase/transcription.

Q2. How is it activated?

➤ Inducer (Lactose/Allo-lactose) binds and inactivates the repressor.

4. Sickle Cell Anaemia Mutation

Q. Type of mutation and amino acid change?

➤ Point mutation; Glutamic acid replaced by Valine at 6th position of beta-globin chain.

5. Genetic Code (Premature Stop Codon)

Q. Effect of UAA/UAG/UGA at 25th position?

➤ Translation terminates early, resulting in a truncated (short) protein.

II. Biotechnology: Principles & Applications

6. Recombinant DNA (EcoRI Cutting)

Q1. Why same enzyme for vector and insert?

➤ To create complementary "sticky ends" for easy ligation.

Q2. Common host?

➤ Escherichia coli (E. coli).

7. PCR (Polymerase Chain Reaction)

Q1. Enzyme used?

➤ Taq polymerase.

Q2. Why heat stable?

➤ Extracted from *Thermus aquaticus*; stays active at high denaturation temps.

8. Bioreactors (Large Scale)

Q. Advantage over lab flasks?

- **Controlled pH, Temp, Oxygen** for mass production of proteins/enzymes.

9. Gene Therapy (ADA Deficiency)

Q. Basic principle?

- **Delivery of a functional gene** to replace/compensate for a non-functional one.

10. Bt Cotton (Pest Resistance)

Q. Source gene and effect?

- **Cry gene from *Bacillus thuringiensis***; creates toxic protein that kills bollworms.

III. Human Reproduction & Health

11. Parturition (Birth Process)

Q. Trigger hormone?

- **Oxytocin** (Fetal ejection reflex triggers its release from posterior pituitary).

12. IVF (In-Vitro Fertilisation)

Q. Where is the 8-celled embryo transferred?

- **Fallopian tube (ZIFT)**; if more than 8 cells, then Uterus (IUT).

13. Sperm Morphology (Acrosome Damage)

Q. Immediate impact?

- **Sperm cannot dissolve Zona Pellucida**; fertilisation fails.

14. Amniocentesis (Legal/Ethical)

Q. Why is it statutorily banned?

- **To prevent misuse for prenatal sex determination and female foeticide.**

15. Down Syndrome (Chromosomal)

Q. Genetic cause?

- **Trisomy of chromosome 21** (Presence of an extra copy).

IV. Flowering Plant Reproduction

16. Double Fertilisation

Q. Fate of the two male gametes?

- **1st + Egg → Zygote (2n); 2nd + 2 Polar Nuclei → PEN (3n).**

17. Endosperm Development

Q. Why does endosperm develop before embryo?

- **To ensure a ready supply of nutrition for the growing embryo.**

18. Hydrophily (Vallisneria)

Q. Method of pollination?

- Female flowers reach surface; pollen grains are released on water surface.

19. Self-Incompatibility

Q. Definition/Advantage?

- Genetic mechanism to prevent self-pollen germination; promotes outbreeding.

20. Geitonogamy

Q. Why is it genetically autogamy?

- Because pollen comes from a different flower but the same individual plant.

V. Evolution, Ecology & Health (Cases 21-50 Summary)

21. Cytochrome-C Homology

Q. SUGGESTION:

- Common ancestry/Molecular evidence for evolution.

22. Industrial Melanism:

- Directional selection favoring dark moths in polluted areas.

23. Analogous Organs (Wings):

- Convergent evolution (Different origin, same function).

24. Fossil Stratigraphy:

- Deeper layers contain older, simpler organisms.

25. Genetic Drift:

- Allele frequency change by chance (e.g., Founder Effect).

26. BOD (Biochemical Oxygen Demand):

- High BOD equals high organic pollution in water.

27. Energy Pyramid:

- Always upright; follows 10% law of energy transfer.

28. Habitat Fragmentation:

- Primary cause of biodiversity loss (Evil Quartet).

29. Mutualism (Fig-Wasp):

- Obligatory relationship; Wasp pollinates, Fig provides egg-laying site.

30. Eutrophication:

- Nutrient enrichment (N, P) leading to Algal bloom and oxygen depletion.

31. HIV/AIDS Pathway:

- Replicates in T-helper (CD4) cells; lowers immunity.

32. Vaccine Principle:

- Artificial active immunity via memory cell formation.

33. Malaria Vector:

- Female Anopheles mosquito (Host for sexual stage).

34. Interferons:

- Produced by virus-infected cells to protect healthy neighbors.

35. Antibiotics vs Viruses:

- Ineffective because viruses lack metabolic/cell-wall pathways.

36. ABO (AB Blood Group):

- Codominance; both Alleles I^A and I^B express equally.

37. Pedigree (Criss-Cross):

- Suggests X-linked recessive inheritance (e.g., Colourblindness).

38. Monohybrid F2 (3:1):

- Proves Law of Segregation (Alleles separate during gamete formation).

39. Linkage vs Crossing Over:

- Linkage keeps genes together; crossing over increases recombination.

40. Pachytene Crossing Over:

- Leads to genetic variation essential for evolution.

41. Biogas (Methanogens):

- Anaerobic bacteria producing Methane (CH_4) from cattle dung.

42. Sewage Flocs:

- Mesh-like masses of aerobic bacteria + fungal filaments.

43. Penicillin Discovery:

- Produced by fungus *Penicillium notatum*; targets bacterial cell walls.

44. Totipotency (Plant Cell):

- The capacity of a cell to regenerate into a whole plant.

45. EtBr (Ethidium Bromide):

- Staining agent used to visualize DNA under UV light in gel electrophoresis.

46. Stop Codons:

- UAA (Ochre), UAG (Amber), UGA (Opal); stop translation.

47. Central Dogma:

- Unidirectional flow of information: $DNA \rightarrow RNA \rightarrow Protein$.

48. Restriction Sticky Ends:

- Overhanging single-stranded DNA sequences produced by EcoRI.

49. Endosperm Nutrition:

- Dense cytoplasm and food reserves used by the embryo.

50. Inverted Number Pyramid:

- Occurs in a single tree ecosystem supporting many insects and parasites.

Top 50 Case Studies By Pragma Thunderstudy • Analyze | Revise | Master